

Stationary Tower Forcing

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Abstract

We introduce the countable stationary tower forcing and show that, under suitable large cardinal assumptions, a small forcing cannot change the theory of Chang's model. These notes are based on [2].

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1 Exposition

1.1 Ultrapowers

- I. Let M be a model of set theory and let U be an M -ultrafilter, i.e. an ultrafilter in $P(X) \cap M$ for some $X \in M$. The *ultrapower* $\text{Ult}(M, U)$ is defined by

$$(\text{Ult}(M, U), \in^U) := (M^X \cap M, =^U, \in^U) / =^U,$$

where

$$\begin{aligned} f =^U g &: \iff \{x \in X : f(x) = g(x)\} \in U, \\ f \in^U g &: \iff \{x \in X : f(x) \in g(x)\} \in U. \end{aligned}$$

- II. **Theorem.** $\text{Ult}(M, U) \models \phi([f] \dots) \iff \{x \in X : M \models \phi(f(x) \dots)\} \in U$

- III. $j_U : M \rightarrow \text{Ult}(M, U)$ is defined by $j(a) := (a : x \in X) / U$. We have $j : M \prec \text{Ult}(M, U)$.

IV. $\text{Ult}(M, U)$ may or may not be well-founded.

V. In these notes, we will be primarily interested in M -ultrafilters that can be found in some generic extension $M[G]$ of M .

VI. Suppose that:

- a. $\mathbb{P}$ is a poset;
- b. $V^{\mathbb{P}} \models \ulcorner \dot{U} \text{ is a } \check{V}\text{-ultrafilter} \urcorner$.

Then $\dot{U}$ is called a *generic ultrafilter* and $\text{Ult}(\check{V}, \dot{U})$ is called a *generic ultrapower* of V .

VII. Suppose that:

- a. $\mathbb{P}$ is a poset;
- b. $V^{\mathbb{P}} \models j : \check{V} \prec M$.

Then j is called a *generic embedding*.

VIII. An example of a generic embedding is the ultrapower embedding $j_{\dot{U}} : V \prec \text{Ult}(\check{V}, \dot{U})$ coming from a generic ultrafilter $\dot{U}$.

1.2 Non-stationary Ideals

I. We will be interested in the non-stationary ideal on

$$[A]^\omega := \{B \subseteq A : |B| \leq \omega\},$$

where A is some set.

II. **Definition.** An *algebra* on A is a mapping $f : A^{<\omega} \rightarrow A$.

III. **Definition.** The *set of closure points* of an algebra f is defined by

$$C_f := \{a \in [A]^\omega : f[a^{<\omega}] \subseteq a\}.$$

IV. **Definition.** The *club filter* club_A on $[A]^\omega$ is the filter on $[A]^\omega$ generated by

$$\{C_f : f : A^{<\omega} \rightarrow A\}.$$

The *non-stationary ideal* NS_A on $[A]^\omega$ is the ideal club_A^* dual to the filter club_A . A set is said to be *club* if it belongs to club_A and it is said to be *stationary* if it is not in NS_A .

V. NS_A (or equivalently, club_A) is ω_1 -complete.

Proof.

1° We may assume that $A \neq \emptyset$.

2° We first verify completeness.

3° Let $(f_n : n < \omega)$ be algebras on A . We want to show that

$$\bigcap_{n < \omega} C_{f_n} \in \text{club}_A.$$

4° Let $(\text{fst}, \text{snd}) : \omega \rightarrow \omega \times \omega$ be some standard coding.

5° We define $f : A^{<\omega} \rightarrow A$ by setting

$$f(x_0 \dots x_{n-1}) := f_{\text{fst}(n)}(x_0 \dots x_{\text{snd}(n)-1}).$$

6° We are done after noting that

$$C_f \subseteq \bigcap_{n < \omega} C_{f_n}.$$

□

VI. Proposition. Let $A \neq \emptyset$ and let $a \subseteq [A]^\omega$. Then the following are equivalent.

- a. $a \in \text{NS}_A^+$
- b. For every countable language $\mathcal{L}$ and every $\mathcal{L}$ -structure $\mathbb{A}$ on A , there is $M \in a$ with $M \prec \mathbb{A}$.

Proof.

1° We first show $a \Rightarrow b$.

2° Given $\mathbb{A}$, let $f : A^{<\omega} \rightarrow A$ code a system of Skolem functions for $\mathbb{A}$.

3° There is $M \in a \cap C_f$.

4° This M is as required.

5° Let us now show $b \Rightarrow a$.

6° Let f be an arbitrary algebra on A . We need to show that $a \cap C_f \neq \emptyset$.

7° There is $M \in a$ such that

$$M \prec (A, f \upharpoonright A^n : n < \omega).$$

8° We have $M \in a \cap C_f$.

□

VII. Remark. In particular, since

$$\{M \prec \mathbb{A} : |M| = \omega\}$$

intersects all the stationary sets, it is club. Also, given any $X \in [A]^\omega$, the set

$$\{M \in [A]^\omega : X \subseteq M\}$$

is club. This follows by considering the structure $(A, x : x \in X)$.

VIII. **Remark.** A consequence is that if a is stationary in $[A]^\omega$, then $A = \cup a$. Thus, we will simply say “ a is stationary” and we shall mean by that “ a is stationary in $[\cup a]^\omega$ ”.

IX. **Proposition.** Let $A \subseteq B$.

a. If $a \in \text{NS}_A^+$, then

$$a \uparrow B := \{M \in [B]^\omega : M \cap A \in a\} \in \text{NS}_B^+.$$

b. If $b \in \text{NS}_B^+$, then

$$b \downarrow A := \{M \cap A : M \in b\} \in \text{NS}_A^+.$$

Proof.

- 1° We first show a.
- 2° Let $\mathbb{B}$ be a structure on B of a countable language.
- 3° We may assume w.l.o.g. that the language of $\mathbb{B}$ contains some system of Skolem functions for $\mathbb{B}$.
- 4° We may treat all the symbols of the language of $\mathbb{B}$ as relational symbols.
- 5° Thus, A is a domain of a substructure $\mathbb{A}$ of $\mathbb{B}$.
- 6° There is $M \in a$ with $M \prec \mathbb{A}$.
- 7° Let N be the definable closure of M in $\mathbb{B}$.
- 8° $N \cap A = M$

Proof.

- 1' Let $x \in N \cap A$.
- 2' Since $x \in N$, we have that $x = f(\bar{y})$ for a Skolem function f and a tuple $\bar{y} \subseteq M$.
- 3' Since $x \in A$, it is defined by the formula $x = f(\bar{y})$ in $\mathbb{A}$ with parameters from M .
- 4' Since $M \prec \mathbb{A}$, we have $x \in M$.

□

- 9° This means $N \in a \uparrow B$.
- 10° We now prove b.
- 11° We may assume w.l.o.g. that $A \neq \emptyset$.
- 12° Let f be an algebra on A .
- 13° Let us fix $x_0 \in A$.
- 14° Let g be an algebra on B defined by setting

$$f(\bar{x}) := \begin{cases} g(\bar{x}), & \bar{x} \in A^{<\omega} \\ x_0, & \text{otherwise.} \end{cases}$$

- 15° Let $M \in b$ be closed for g .
 16° Then $M \cap A \in b \downarrow A$ is closed for f .

□

X. The non-stationary ideals have an additional important property, namely *normality*.

XI. **Definition.** Let M be a model of set theory and let I be an M -ideal on $([A]^\omega)^M$. Then I is said to be *M -normal* if for every f , the conditions:

- a. $S \in (P([A]^\omega))^M - I$;
- b. $f : S \rightarrow A$;
- c. $f \in M$;
- d. $(\forall a \in S)(f(a) \in a)$;

imply

$$(\exists T \in P(S)^M - I)(f \upharpoonright T = \text{const.}).$$

XII. **Remark.** By simply “normal”, we shall mean V -normal.

XIII. **Definition.** For $f : A \rightarrow P([A]^\omega)$, we define

$$\Delta_{a \in A} f(a) := \{b \in [A]^\omega : b \in \bigcap_{a \in b} f(a)\} \subseteq [A]^\omega,$$

$$\nabla_{a \in A} f(a) := \{b \in [A]^\omega : b \in \bigcup_{a \in b} f(a)\} \subseteq [A]^\omega.$$

XIV. **Proposition.** An M -ideal I on $([A]^\omega)^M$ is M -normal if and only if it is closed for ∇^M , i.e. if for every $f : A \rightarrow I$ with $f \in M$, we have $\nabla f \in I$. This is also equivalent to I^* being closed for Δ^M .

XV. **Proposition.** NS_A is normal.

Proof.

- 1° We check the closure for Δ .
 2° Let $(f_x : x \in A)$ be algebras on A . We want to show that

$$\Delta_{x \in A} C_{f_x} \in \text{club}_A.$$

- 3° Let f be an algebra on A defined by

$$f(x_0 \dots x_n) := f_{x_0}(x_1 \dots x_n)$$

($f(\emptyset)$ can be anything).

- 4° We are done after noting that

$$C_f \subseteq \Delta_{x \in A} C_{f_x}.$$

□

2 Stationary Tower

I. **Definition.** Let $\delta > \omega$ be a limit ordinal. The (countable) stationary tower $\mathbb{Q}_\delta$ is defined as follows.

- a. Its domain is the set of all stationary $a \in V_\delta$.
- b. $a \leq_{\mathbb{Q}_\delta} b$ if and only if
 - i. $\cup a \supseteq \cup b$,
 - ii. $a \subseteq b \upharpoonright \cup a$.

II. **Proposition.** Let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ and let

$$G_A := \{a \in G : \cup a = A\} \subseteq P^V(A)$$

for $A \in V_\delta$. Then:

- a. G_A is a V -ultrafilter;
- b. G_A extends club_A^V .

Proof.

Part a.

- 1° We only need to verify that for every $a \in (P([A]^\omega))^V$, either $a \in G_A$ or $a^{\mathbb{G}} \in G_A$.
- 2° For this, it suffices to show that $\{a, a^{\mathbb{G}}\}$ is predense in $\mathbb{Q}_\delta$.
- 3° Let $b \in \mathbb{Q}_\delta$ be arbitrary.
- 4° Let $C := (\cup a) \cup (\cup b)$. We have

$$b \upharpoonright C = ((a \upharpoonright C) \cap (b \upharpoonright C)) \cup ((a^{\mathbb{G}} \upharpoonright C) \cap (b \upharpoonright C)).$$

- 5° Since $b \upharpoonright C$ is stationary, at least one of the sets

$$((a \upharpoonright C) \cap (b \upharpoonright C)), \quad ((a^{\mathbb{G}} \upharpoonright C) \cap (b \upharpoonright C))$$

is stationary, witnessing either $a \parallel b$ or $a^{\mathbb{G}} \upharpoonright b$.

Part b.

- 6° This follows from the fact that every club set in $\mathbb{Q}_\delta$ is equal to 1 in $\mathbb{B}(\mathbb{Q}_\delta)$.

□

III. Thus, G can be understood as a family $(G_A : A \in V_\delta)$ of generic¹ ultrafilters. This system is coherent in the following sense.

IV. Let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ and let $A \subseteq B \in V_\delta$.

¹“Generic” in the sense of definition 1.1-VI.

- a. If $a \in G_A$, then $a \uparrow B \in G_B$.
- b. If $b \in G_B$, then $b \downarrow A \in G_A$.

Proof.

- 1° We first show a.
- 2° Let $a \in G \cap P([A]^\omega)$ be arbitrary. We want to show that $a \uparrow B \in G$.
- 3° It suffices to show that $\{a \uparrow B\}$ is predense below a .
- 4° Let $c \leq a$ be arbitrary.
- 5° $\cup c \supseteq A$
- 6° $c \subseteq a \uparrow \cup c$
- 7° $c \uparrow ((\cup c) \cup B) \subseteq (a \uparrow \cup c) \uparrow ((\cup c) \cup B) = a \uparrow ((\cup c) \cup B) = (a \uparrow B) \uparrow ((\cup c) \cup B)$
- 8° Thus, $c \uparrow ((\cup c) \cup B)$ witnesses $c \parallel a \uparrow B$.
- 9° Regarding b, it is obvious since $b \leq b \downarrow A$.

□

V. Let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ and let $A \in V_\delta$. Then G_A is V -normal.

Proof.

- 1° Let $a \in G_A$ and let $f : a \rightarrow A$ be a choice function in V . We want to find $a' \in G_A$ with $a' \subseteq a$ such that $f \upharpoonright a'$ is constant.
- 2° For $B \supseteq A$, let us denote $f^B : a \uparrow B \rightarrow A : M \mapsto f(M \cap A)$. It suffices to find some $a' \in G$ with $a' \leq a$ such that $f^{\cup a'} \upharpoonright a' = \text{const.}$
- 3° For this, in turn, it suffices to show that $D := \{b \leq a : f^{\cup b} \upharpoonright b = \text{const.}\}$ is predense below a .
- 4° Let $a' \leq a$ be arbitrary.
- 5° $f^{\cup a'} \upharpoonright a' : a' \rightarrow A \subseteq \cup a'$ is a choice function on a stationary set a' .
- 6° Since $\text{NS}_{\cup a'}$ is normal in V , there is $b \in \text{NS}_{\cup a'}^+$ with $b \subseteq a'$ and $f^{\cup a'} \upharpoonright b = (f^{\cup a'} \upharpoonright a') \upharpoonright b = \text{const.}$
- 7° Thus, $b \leq a'$ and $b \in D$.

□

VI. For $A \subseteq B \in V_\delta$, if $f : [A]^\omega \rightarrow V$, we define its lifting

$$f^B : [B]^\omega \rightarrow V : N \mapsto f(N \cap A).$$

We then define

$$j_{A,B}^G : \text{Ult}(V, G_A) \prec \text{Ult}(V, G_B) : [f]_{G_A} \mapsto [f^B]_{G_B}.$$

Of course, one needs to verify that $j_{A,B}^G$ is correctly defined and indeed elementary, but this simply follows from 1.1-II and 2-IV.

VII. It is a matter of routine to verify that

$$(\text{Ult}(V, G_A), j_{A,B}^G : A \subseteq B \in V_\delta)$$

is a directed system of models and embeddings. We denote its direct limit by $(\text{Ult}(V, G), \in^G)$ and the corresponding embeddings by $j_{A,\infty}^G$ ($A \in V_\delta$). In particular, we have $j_G := j_{0,\infty}^G : V \prec \text{Ult}(V, G)$.

VIII. We also have $(\text{Ult}(V, G), \in^G) \cong (D, =^D, \in^D) / =^D$, where

$$D := \{f : [A]^\omega \rightarrow V : A \in V_\delta, f \in V\},$$

while for $A, B \in V_\delta$ and $f : [A]^\omega \rightarrow V$ and $g : [B]^\omega \rightarrow V$,

$$f =^D g : \iff \{f(P \cap A) = g(P \cap B) : P \in [A \cup B]^\omega\} \in G,$$

$$f \in^D g : \iff \{f(P \cap A) \in g(P \cap B) : P \in [A \cup B]^\omega\} \in G.$$

IX. **Theorem.** For $A_i \in V_\delta$, $f_i : [A_i]^\omega \rightarrow V$ with $f_i \in V$, and $C \in V_\delta$ with $C \supseteq A_0 \cup \dots \cup A_{n-1}$, we have

$$\text{Ult}(V, G) \models \phi([f_0] \dots [f_{n-1}]) \iff$$

$$\{P \in [C]^\omega : V \models \phi(f_0(P \cap A_0) \dots f_{n-1}(P \cap A_{n-1}))\} \in G.$$

Proof. Routine. □

X. **Remark.** When considering models (B, E) of set theory, we say that $u \in B$ is in its well-founded part (denoted by $\text{wfp}(B, E)$) if $(\{x \in B : xE^*u\}, E)$ is a well-founded structure. We have that $(\text{wfp}(B, E), E)$ is itself a well-founded structure, so it is identified with its transitive collapse. Thus, $\alpha := \text{wfp}(B, E) \cap \text{Ord}$ is an ordinal and $\alpha \subseteq B$. Moreover, for $u \in B$, we have

$$u \in \text{wfp}(B, E) \iff \text{rank}^{(B, E)}(u) < \alpha.$$

XI. **Proposition.** Let $\delta \geq \omega_1$ be a limit ordinal. Let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$, let $\mathbf{M} := \text{Ult}(V, G)$, and let $j := j_G : V \prec \mathbf{M}$. Then the following holds.

- a. For every $A \in V_\delta$, we have $j[A] \text{ “} \in \text{” } \mathbf{M}$, in the sense that: there is $a \in \mathbf{M}$ such that $\{x \in \mathbf{M} : x \in^G a\} = j[A]$. In fact, $a = [\text{id}_{[A]^\omega}]$.
- b. For every $a \in V_\delta$, we have that $a \in G$ if and only if $j[\cup a] \in^G j(a)$.
- c. $\delta \subseteq \text{wfp}(\mathbf{M})$
- d. $\text{crit}(j) = \omega_1^V$
- e. $j(\omega_1^V) \text{ “} \geq \text{” } \delta$, in the sense that²: for every $\alpha < \delta$, we have $j(\omega_1^V) > \alpha$.

²Note that we do not know if $\delta \in \mathbf{M}$.

Proof.

- 1° We first show [a](#).
- 2° Let $x \in A$ be arbitrary. We have $j(x) \in [\text{id}_{[A]^\omega}]$, since $\{M \in [A]^\omega : x \in M\}$ is a club and hence $\{M \in [A]^\omega : x \in M\} \in G$.
- 3° Thus, $j[A] \subseteq [\text{id}_{[A]^\omega}]$.
- 4° On the other hand, let $[f] \in^G [\text{id}_{[A]^\omega}]$ be arbitrary.
- 5° We may assume that $f : [B]^\omega \rightarrow V$ for some $B \supseteq A$.
- 6° We have

$$G \ni \{M \in [B]^\omega : f(M) \in \text{id}_{[A]^\omega}^B(M)\} = \{M \in [B]^\omega : f(M) \in M \cap A\} =: b.$$

- 7° The normality of G_B now implies the existence of some $b' \in G$ with $b' \subseteq b$ and some $x \in A$ such that

$$(\forall M \in b')(f(M) = x).$$

- 8° This means $[f] = [(x : M \in [B]^\omega)] = j(x) \in j[A]$.
- 9° [b](#) follows immediately from [a](#).
- 10° Let us now consider [c](#). Let $\alpha < \delta$ be arbitrary.
- 11° We have $j[\alpha] \in^M \mathbf{M}$.
- 12° Since α is computed as the order type of $(j[\alpha], \in^G)$, both in $V[G]$ and in $\mathbf{M}$, we conclude $\alpha \in \text{wfp}(\mathbf{M})$.
- 13° We will now show $\text{crit } j \geq \omega_1^V$. Property [d](#) will then follow once we establish [e](#).
- 14° Let $\alpha < \omega_1^V$. We want to show $j(\alpha) = \alpha$.
- 15° We have $\alpha = \text{ot}^{\mathbf{M}}(j[\alpha]) = \text{ot}^{\mathbf{M}}([\text{id}_{[A]^\omega}]) = [(\text{ot}(M) : M \in [A]^\omega)]$.
- 16° Since $|\alpha|^V < \omega_1^V$, we have that $\{\alpha\} = \{M \in [A]^\omega : M \supseteq \alpha\}$ is club in $[A]^\omega$.
- 17° This means $\alpha = [(\text{ot}(M) : M \in [A]^\omega)] = [(\alpha : M \in [A]^\omega)] = j(\alpha)$.
- 18° We now show [e](#).
- 19° Let $\alpha < \delta$. We want to show that $\alpha < j(\omega_1^V)$.
- 20° We again have $\alpha = [(\text{ot}(M) : M \in [A]^\omega)]$.
- 21° Hence, the conclusion follows from

$$\{M \in [A]^\omega : \text{ot}(M) < \omega_1^V\} = [A]^\omega \in G.$$

□

XII. We obtain additional properties of j_G by imposing large cardinal properties on δ . We start by obtaining $j(\omega_1^V) = \delta$.

XIII. **Definition.** An uncountable cardinal κ is *HEP*³ if for every $a \in \mathbb{Q}_\kappa$ and every $f : a \rightarrow \omega_1$, the set

$$\{M \in a \uparrow V_\kappa : \text{ot}(M \cap \kappa) \geq f(M \cap (\cup a))\}$$

is stationary in $[V_\kappa]^\omega$.

XIV. If δ is a limit of HEP cardinals, then

$$V^{\mathbb{Q}_\delta} \models j_{\dot{G}}(\omega_1^{\dot{V}}) = \delta$$

(where $\dot{G}$ is the canonical name for the generic).

Proof.

1° Keeping in mind 2-XI.e, it suffices to show that for every $A \in V_\delta$ and $f : [A]^\omega \rightarrow \omega_1$, we have

$$V^{\mathbb{Q}_\delta} \models (\exists \alpha < \delta)([f] \leq \alpha).$$

2° Suppose that:

- a. $A \in V_\delta$
- b. $f : [A]^\omega \rightarrow \omega_1$
- c. $a \in \mathbb{Q}_\delta$.

We want to find $b \leq a$ and $\alpha < \delta$ such that $b \Vdash [f] \leq \alpha$.

3° Up to passing to $f^{A \cup (\cup a)}$ and $a \uparrow (A \cup (\cup a))$, we may assume that $\cup a = A$.

4° Let $\alpha < \delta$ be a HEP cardinal such that $A \in V_\alpha$.

5° Since $a \in \mathbb{Q}_\alpha$, we have

$$b := \{M \in a \uparrow V_\alpha : \text{ot}(M \cap \alpha) \geq f(M \cap A)\} \in \mathbb{Q}_\delta.$$

6° Obviously, $b \leq a$ and $b \Vdash [f] \leq \alpha$.

□

XV. A natural question is, of course, which cardinals are limits of HEP cardinals. We will show that measurable and Woodin cardinals are two examples of such cardinals.

XVI. **Proposition.** The following holds.

- a. Every measurable cardinal is HEP.
- b. Every measurable cardinal is a limit of HEP cardinals.
- c. Every Woodin cardinal is a limit of HEP cardinals.

³This is not a standard notion. The acronym stands for “height extension property”.

Proof.

- 1° Let κ be measurable and let us show that κ is HEP.
- 2° Let $a \in \mathbb{Q}_\kappa$ and $F : V_\kappa^{<\omega} \rightarrow V_\kappa$ be arbitrary. We want to find N such that
 - a. $N \in [V_\kappa]^\omega$
 - b. $F[N^{<\omega}] \subseteq N$
 - c. $N \cap (\cup a) \in a$
 - d. $\text{ot}(N \cap \kappa) \geq f(N \cap (\cup a))$.
- 3° Let μ be a normal measure on κ and let $\lambda := (2^{2^\kappa})^+$.
- 4° Since a is stationary, there is a countable

$$M' \prec (H_\lambda, \in, \mu, F, a)$$

such that $M' \cap (\cup a) \in a$.

- 5° Let M be the collapse of M' and let

$$k_0 : (M, \in, \bar{\kappa}, \bar{\mu}) \prec (H_\lambda, \in, \kappa, \mu)$$

be the inverse of the collapsing map.

- 6° Let

$$(M_\alpha, j_{\alpha,\beta} : \alpha < \beta < \omega_1)$$

be the ω_1 -iteration of M via $\bar{\mu}$.

- 7° This iteration is realizable in H_λ , in the following sense. There are $k_\alpha : M_\alpha \prec H_\lambda$ ($\alpha < \omega_1$) such that

$$\begin{array}{ccc} & H_\lambda & \\ k_\alpha \uparrow & \nwarrow k_\beta & \\ M_\alpha & \xrightarrow{j_{\alpha,\beta}} & M_\beta \end{array}$$

commutes for every $\alpha < \beta < \omega_1$.

- 8° Setting $\alpha := f(M' \cap (\cup a))$, we have $N' := k_\alpha[M_\alpha]$ and $N := N' \cap V_\kappa$.
- 9° Let us show that N is as required in 2°.
- 10° N is countable, since M_α is countable.
- 11° Since $F \in M' \subseteq N'$, we have that N' , and consequently N , are closed for F .
- 12° Since $\text{crit } j_{0,\alpha} = \bar{\kappa}$, we have that $k_0[M_0] = M'$ and $k_\alpha[M_\alpha] = N'$ agree up to $\sup(\kappa \cap M_0)$.
- 13° In particular, $N \cap (\cup a) = N' \cap (\cup a) = M' \cap (\cup a) \in a$.
- 14° Finally,

$$\begin{aligned} \text{ot}(N \cap \kappa) &= \text{ot}(N' \cap \kappa) = \text{ot}(M_\alpha \cap j_{0,\alpha}(\bar{\kappa})) \geq \\ &\text{ot}(\{j_{0,\beta}(\bar{\kappa}) : \beta < \alpha\}) = \alpha = f(M' \cap (\cup a)) = f(N' \cap (\cup a)). \end{aligned}$$

15° Let us now consider the part [b](#).

16° Since κ is HEP, it is easily seen that

$$\text{Ult}(V, \mu) \models \ulcorner \kappa \text{ is HEP} \urcorner.$$

17° By normality, the set of HEP cardinals below κ has μ measure 1.

18° Thus, κ is a limit of HEP cardinals.

19° Part [c](#) is immediate from [a](#) and the fact that a Woodin cardinal is a limit of measurable cardinals.

□

3 The Case of a Woodin Cardinal

I. We introduce an auxiliary property that we call “semi-proper reflection”. We show that Woodin cardinals have this property. We use that property to deduce some consequences, in particular that the stationary tower ultrapower is well-founded and closed for countable sequences.

II. **Definition.** Suppose that κ is inaccessible and $D \subseteq \mathbb{Q}_\kappa$. Then the set⁴ $\text{capt_ext}(D) \subseteq [H(\kappa^+)]^\omega$ consists exactly of all countable $M \prec (H(\kappa^+), \in, D)$ for which there exists a countable $N \prec H(\kappa^+)$ satisfying:

- a. $M \subseteq N$
- b. $N \cap V_\kappa$ end-extends⁵ $M \cap V_\kappa$;
- c. there is $d \in N \cap D$ such that $N \cap (\cup d) \in d$.

III. **Definition.** $D \subseteq \mathbb{Q}_\kappa$ is *semi-proper* if it is predense and $\text{capt_ext}(D)$ is club in $[H(\kappa^+)]^\omega$.

IV. **Lemma.** Let $D \subseteq \mathbb{Q}_\kappa$ be predense and let $\theta > 2^\kappa$ be regular. The the following are equivalent:

- a. D is semi-proper;
- b. for every⁶ countable $M \prec (H_\theta, \in, D, \kappa)$, there is a countable $N \prec H_\theta$ such that
 - i. $M \subseteq N$
 - ii. $N \cap V_\kappa$ end-extends $M \cap V_\kappa$;
 - iii. N captures D .

Proof. We first prove a pre-lemma that we need and then we give a proof of the proposition in [3-VI](#). □

⁴We want to say “the set of all M that can be extended to capture D ”.

⁵This means that $N \cap V_\kappa$ and $M \cap V_\kappa$ agree up to $\sup(M \cap \kappa)$.

⁶What we get by passing to larger H_θ is the possibility to extend all submodels M , instead of just club-many of them.

V. **Pre-lemma.** Suppose that:

- a. $\theta > \omega$ is regular;
- b. $M \prec H_\theta$
- c. $A \in M$
- d. $X \subseteq A$
- e. $M[X] := \{f(\bar{x}) : f : A^{<\omega} \rightarrow H_\theta, f \in M, \bar{x} \in X^{<\omega}\}.$

Then:

- i $M[X] \prec H_\theta$
- ii $(\forall N \prec H_\theta)(M \cup X \subseteq N \implies M[X] \prec N)$
- iii $|M[X]| = \max\{|M|, |X|\}.$

Proof.

- 1° For i, we apply Tarski-Vaught test. Suppose that $H_\theta \models \exists y \phi(f(x), y)$, for some $f \in H_\theta^A \cap M$ and $x \in X$.
- 2° $Y := \{s \in A : \exists y \phi(f(s), y)\}^{H_\theta} = \{s \in A : \exists y \phi(f(s), y)\}^M \in M$
- 3° Let $g : Y \rightarrow H_\theta$ satisfy $g \in M$ and

$$M \models (\forall s \in Y) \phi(f(s), g(s)).$$

- 4° $H_\theta \models (\forall s \in Y) \phi(f(s), g(s))$
- 5° Since $x \in Y$, we have $H_\theta \models \phi(f(x), g(x))$, while $g(x) \in M[X]$.
- 6° ii and iii are obvious.

□

VI. *Proof of 3-IV.*

Implication (a $\Rightarrow$ b)

- 1° Let $M \prec (H_\theta, \in, D, \kappa)$ be arbitrary countable.

- 2° Since

$$H_\theta \models \text{capt_ext}(D) \in \text{club}_{H(\kappa^+)},$$

we have

$$M \models \text{capt_ext}(D) \in \text{club}_{H(\kappa^+)}.$$

- 3° Thus, there is $F : H(\kappa^+)^{<\omega} \rightarrow H(\kappa^+)$ with $F \in M$ and $C_F \subseteq \text{capt_ext}(D)$.
- 4° Since $F \in M$, we have that $M \cap H(\kappa^+) \in C_F$ and consequently, $M \cap H(\kappa^+) \in \text{capt_ext}(D)$.
- 5° Thus, there is a countable $M' \prec H(\kappa^+)$ such that:
 - a. $M \cap H(\kappa^+) \subseteq M'$;

- b. $M' \cap V_\kappa$ end-extends $M \cap V_\kappa$;
 - c. M' captures D .
- 6° Since $M' \cap V_\kappa \subseteq V_\kappa \in M$, we have

$$N := M[M' \cap V_\kappa] \prec H_\theta.$$

7° $M \subseteq N$

8° **Claim.** $N \cap V_\kappa = M' \cap V_\kappa$

Proof.

- 1' Let $f : V_\kappa \rightarrow H_\theta$ with $f \in M$ and $x \in M' \cap V_\kappa$ be such that $f(x) \in V_\kappa$.
We want to show that $f(x) \in M'$.
- 2' Let $g : V_\kappa \rightarrow V_\kappa$ be defined by

$$g(y) := \begin{cases} f(y), & f(y) \in V_\kappa \\ 0, & \text{otherwise.} \end{cases}$$

- 3' We have $g \in M \cap H(\kappa^+) \subseteq M'$.
- 4' Thus, $f(x) = g(x) \in M'$.

□

- 9° Then $N \cap V_\kappa = M' \cap V_\kappa$ end-extends $M \cap V_\kappa$.
- 10° Since $\text{capt}(M', D)$, there is $d \in M' \cap D$ such that $M' \cap (\cup d) \in d$.
- 11° $M' \cap D = (M' \cap V_\kappa) \cap D = (N \cap V_\kappa) \cap D = N \cap D$
- 12° Similarly, $N \cap (\cup d) = M' \cap (\cup d)$.
- 13° Thus, d witnesses $\text{capt}(N, D)$.
- Implication (b $\Rightarrow$ a)**
- 14° Assume towards contradiction that $\text{capt_ext}(D) \subseteq [H(\kappa^+)]^\omega$ is not club.
- 15° This means that $\text{capt_ext}(D)^\mathbb{C}$ is stationary.
- 16° So $\text{capt_ext}(D)^\mathbb{C} \uparrow H_\theta$ is stationary.
- 17° Hence, there is a countable $M \prec (H_\theta, \in, D, \kappa)$ with $M \cap H(\kappa^+) \notin \text{capt_ext}(D)$.
- 18° By b, there is a countable $N \prec H_\theta$ such that

- $M \subseteq N$,
- $N \cap V_\kappa$ end-extends $M \cap V_\kappa$,
- N captures D .

- 19° Then, $N \cap H(\kappa^+)$ witnesses that $M \cap H(\kappa^+) \in \text{capt_ext}(D)$, which is a contradiction.

□

VII. **Definition.** δ is said to have *semi-proper reflection property* if it is inaccessible and for every sequence $(D_\xi : \xi < \delta)$ of predense subsets of $\mathbb{Q}_\delta$, there is some inaccessible $\kappa < \delta$ satisfying

$$(\forall \xi < \kappa) \text{semi_proper}(D_\xi \cap \mathbb{Q}_\kappa).$$

VIII. **Remark.** A witness κ for the above property can be chosen to be arbitrarily large. Indeed, given some $\gamma < \delta$, one can consider an additional predense set $D_{-\gamma} := \mathbb{Q}_\delta - V_\gamma$ and apply the property. A witness obtained this way satisfies $\kappa > \gamma$.

IX. **Theorem.** If δ is Woodin, it satisfies the semi-proper reflection property.

Proof.

- 1° Let $(D_\alpha : \alpha < \delta)$ be an arbitrary family of predense subsets of $\mathbb{Q}_\delta$.
- 2° Since all D_α 's are predense, there is an increasing $f : \delta \rightarrow \delta$ satisfying

$$(\forall \alpha < \delta)(\forall a \in \mathbb{Q}_\delta \cap V_\alpha)(\forall \beta < \alpha)(\exists b \in D_\beta \cap V_{f(\alpha)})(a \parallel b).$$

- 3° Note that if $\kappa < \delta$ is inaccessible and closed for f , then $D_\alpha \cap \mathbb{Q}_\kappa$ is predense in $\mathbb{Q}_\kappa$.
- 4° **Claim.** There are κ, λ, E such that:
 - a. λ is inaccessible;
 - b. E is a strong (κ, λ) -extender;
 - c. κ, λ are closed for f ;
 - d. $j_E(f) \restriction \lambda = f \restriction \lambda$.

Proof.

- 1' Let $\kappa < \delta$ satisfy: for every $\lambda \in (\kappa, \delta)$, there is a strong (κ, λ) -extender E such that $j_E(f) \cap V_\lambda = f \cap V_\lambda$.
- 2' Let $\lambda \in (\kappa, \delta)$ be closed for f and inaccessible.
- 3' Let E be a strong (κ, λ) -extender such that $j_E(f) \cap V_\lambda = f \cap V_\lambda$.
- 4' κ, λ, E are as required.

□

- 5° We want to show that κ has the property that for every $\alpha < \kappa$, the set $D_\alpha \cap \mathbb{Q}_\kappa$ is semi-proper.
- 6° To that end, fix some $\alpha < \kappa$ and assume towards contradiction that $D_\alpha \cap \mathbb{Q}_\kappa$ is not semi-proper.
- 7° Since $f[\kappa] \subseteq \kappa$, we have that $D_\alpha \cap \mathbb{Q}_\kappa$ is predense in $\mathbb{Q}_\kappa$, meaning that $\text{capt_ext}(D_\alpha \cap \mathbb{Q}_\kappa)$ fails to be club in $H(\kappa^+)$.
- 8° Thus, $a := \text{capt_ext}(D_\alpha \cap \mathbb{Q}_\kappa)^\mathbb{G} \in \mathbb{Q}_\delta \cap V_\lambda$.
- 9° Let us denote $j := j_E$ and $M := \text{Ult}(V, E)$.

10° Keeping in mind that $M \upharpoonright \lambda = V_\lambda$ and that j is elementary, we reason in M .

1' $j(f)[\lambda] \subseteq \lambda$

2' Thus, $j(D_\alpha) \cap \mathbb{Q}_\lambda$ is predense in $\mathbb{Q}_\lambda$.

3' $a \in \mathbb{Q}_\delta \cap V_\lambda = \mathbb{Q}_\lambda$, so there is some $b \in j(D_\alpha) \cap \mathbb{Q}_\lambda$ compatible with a .

4' There is $c \in \mathbb{Q}_\lambda = \mathbb{Q}_\delta \cap V_\lambda$ such that $c \leq a, b$.

11° In V , we also have: $a, b, c \in \mathbb{Q}_\delta \cap V_\lambda$ and $c \leq a, b$.

12° Let $H \prec (H_\delta, \in, \kappa, E, D_\alpha, b)$ be countable with $H \cap (\cup c) \in c$.

13° **Claim.** $M \models j(H \cap H(\kappa^+)) \in j(\text{capt_ext}(D_\alpha \cap \mathbb{Q}_\kappa))$

Proof.

1' In other words, we need to show that

$$M \models j(H \cap H(\kappa^+)) \in \text{capt_ext}(j(D_\alpha) \cap \mathbb{Q}_{j(\kappa)}).$$

2' Since $E \in H$, we have $j(H \cap H(\kappa^+)) = j[H \cap H(\kappa^+)] \subseteq H$.

3' Since M is ω -closed, we have $H \cap H(j(\kappa)^+)^M \in M$.

4' We argue that $H \cap H(j(\kappa)^+)^M$ witnesses [13°1'](#).

5' $j(H \cap H(\kappa^+)) \subseteq H \cap H(j(\kappa)^+)^M$ follows from [13°2'](#).

6' Since

$$j(H \cap H(\kappa^+)) \cap (M \upharpoonright j(\kappa)) = j[H \cap H(\kappa^+)] \cap j(V_\kappa) = j[H \cap V_\kappa] = H \cap V_\kappa,$$

we conclude that $H \cap H(j(\kappa)^+)^M$ end-extends $j(H \cap H(\kappa^+))$.

7' Since $H \cap (\cup c) \in c$ and $c \leq b$, we have $H \cap (\cup b) \in b$.

8' Since $b \in M \upharpoonright \lambda$, we conclude

$$(H \cap H(j(\kappa)^+)^M) \cap (\cup b) \in b.$$

9' By [10°3'](#) and [12°](#), we have

$$b \in (j(D_\alpha) \cap \mathbb{Q}_{j(\kappa)}^M) \cap (H \cap H(j(\kappa)^+)^M).$$

10' Thus, b witnesses that $H \cap H(j(\kappa)^+)^M$ captures $j(D_\alpha) \cap \mathbb{Q}_{j(\kappa)}^M$.

11' [13°5'](#), [13°6'](#), and [13°10'](#) are what is required. □

14° The claim and the elementarity of j imply

$$H \cap H(\kappa^+) \in \text{capt_ext}(D_\alpha \cap \mathbb{Q}_\kappa) = a^{\mathbb{G}}.$$

15° However, $H \cap (\cup c) \in c$ and $c \leq a$ imply $H \cap H(\kappa^+) \in a$, so we have reached contradiction. □

X. **Definition.** Let δ be inaccessible and $\theta > \delta$ regular. Model $M \prec (H_\theta, \in, \delta)$ is said to be *self-generic* if it captures every predense subset of $\mathbb{Q}_\delta$ that it contains.

XI. **Remark.** Instead of considering $M \prec (H_\theta, \in, \delta)$, we can consider the corresponding anti-collapse $k : (\overline{M}, \in, \bar{\kappa}) \prec (H_\theta, \in, \kappa)$. We can derive from it a filter G_k on $\mathbb{Q}_{\bar{\kappa}}^{\overline{M}}$ by setting

$$a \in G_k : \iff k[\cup a] \in k(a)$$

for $a \in \mathbb{Q}_{\bar{\kappa}}^{\overline{M}}$. Self-genericity now amounts to saying that G_k is $\mathbb{Q}_{\bar{\kappa}}^{\overline{M}}$ -generic over M .

XII. **Lemma.** Suppose that:

- a. δ is Woodin;
- b. $\alpha < \delta$
- c. $\theta > \delta$ is regular;
- d. $M \prec (H_\theta, \in, \delta, \alpha)$ and $|M| = \omega$.

Then there is a countable $N \prec H_\theta$ such that

- i. $M \subseteq N$,
- ii. $M \cap V_\alpha = N \cap V_\alpha$,
- iii. N is self-generic.

Proof.

1° Let $(\pi_0, \pi_1) : \omega \rightarrow \omega \times \omega$ be the usual pairing function.

2° Objects $M_i, d_i, (D_j^i : j < \omega)$ are defined by recursion on $i < \omega$, satisfying:

- a. $M_0 := M$
- b. $M_i \prec H_\theta$ and $|M_i| = \omega$;
- c. $M_i \subseteq M_{i+1}$;
- d. M_i and M_{i+1} agree up to $\alpha \cup \bigcup_{j < i} \text{rank}(d_j)$;
- e. $(D_j^i : j < \omega)$ enumerates predense subsets of $\mathbb{Q}_\delta$ that are in M_i ;
- f. d_i witnesses that M_{i+1} captures $D_{\pi_1(i)}^{\pi_0(i)}$.

3° *Recursion.*

- 1' The condition 2°a settles the base case.
- 2' Assuming that the recursion has been completed up to stage i , we proceed to settle the stage $i + 1$.
- 3' Since δ is Woodin, there is an inaccessible $\kappa \in (\alpha \cup \bigcup_{j < i} \text{rank}(d_j), \delta)$ such that $D_{\pi_1(i)}^{\pi_0(i)} \cap \mathbb{Q}_\kappa$ is semi-proper. By elementarity, κ can be chosen in M_i .
- 4' Since $D_{\pi_1(i)}^{\pi_0(i)} \cap \mathbb{Q}_\kappa$ is semi-proper, lemma 3-IV yields M_{i+1} and d_i such that

- a. $M_{i+1} \prec H_\theta$ and $|M_{i+1}| = \omega$,
 - b. $M_i \subseteq M_{i+1}$,
 - c. $M_{i+1} \cap V_\kappa$ end-extends $M_i \cap V_\kappa$,
 - d. $d_i \in D_{\pi_1(i)}^{\pi_0(i)} \cap \mathbb{Q}_\kappa \cap M_{i+1}$ and $M_{i+1} \cap \bigcup d_i \in d_i$.
- 5' Since $\kappa > \alpha \cap \bigcup_{j < i} \text{rank}(d_i) \in M_i$, property 3°4'c implies that M_i and M_{i+1} agree up to $\alpha \cup \bigcup_{j < i} \text{rank}(d_j)$.
- 6' Let $(D_j^{i+1} : j < \omega)$ enumerate predense subsets of $\mathbb{Q}_\delta$ that are in M_{i+1} .
- 7' This completes the recursion.

□

4° Let $N := \bigcup_{i < \omega} M_i$.

5° N is as required.

□

XIII. **Lemma.** Let δ be Woodin and let

$$a_\delta := \{M \prec H(\delta^+) : |M| = \omega \wedge M \text{ is self-generic}\}.$$

Then a_δ is stationary. Moreover, a_δ is *projectively stationary*, i.e. for every $b \in \mathbb{Q}_\delta$, the intersection $a_\delta \cap (b \upharpoonright H(\delta^+))$ is stationary.

Proof.

1° Let $F : H(\delta^+)^{<\omega} \rightarrow H(\delta^+)$ and let $b \in \mathbb{Q}_\delta$ be arbitrary. We want to find $N \in a_\delta$ with $N \cap (\cup b) \in b$ and closed for F .

2° Let $\theta := (2^\delta)^+$ and let

$$M \prec (H_\theta, \in, \delta, F)$$

be countable with $M \cap (\cup b) \in b$.

3° By lemma 3-XII, there is a countable $N \prec H_\theta$ extending M and capturing all the predense subsets of $\mathbb{Q}_\delta$ that it contains.

4° Then $N \cap H(\delta^+)$ is as required.

□

XIV. **Proposition.** If δ is Woodin and $G \rightsquigarrow V^{\mathbb{Q}_\delta}$, then $\text{Ult}(V, G)$ is well-founded.

Proof.

1° We will only use the fact that a_δ is projectively stationary.

2° Assume otherwise.

3° Then there is $a \in \mathbb{Q}_\delta$ such that $a \Vdash \ulcorner \text{Ult}(\check{V}, \Gamma) \text{ is ill-founded.} \urcorner$ (where Γ is the canonical name for the generic).

4° Then for some large enough regular θ , we have

$$a \Vdash \ulcorner \text{Ult}(H_\theta^{\check{V}}, \Gamma) \text{ is ill-founded} \urcorner.$$

5° Since a_δ is projectively stationary, there is a countable $M' \prec (H_\theta, \in, \delta, a)$ with $M' \cap H(\delta^+) \in a_\delta$ and $M' \cap (\cup a) \in a$.

6° Let $k : (M, \in, \bar{\delta}, \bar{a}) \prec (H_\theta, \in, \delta, a)$ be the anti-collapse mapping for M' .

7° k is self-generic.

8° Thus, $G_k \rightsquigarrow M^{\mathbb{Q}_\delta^M}$ and $G_k \ni \bar{a}$.

9° Since by elementarity

$$\bar{a} \Vdash \ulcorner \text{Ult}(\text{Ground}, \text{generic}) \text{ is ill-founded} \urcorner,$$

we have that $\text{Ult}(M, G_k)$ is ill-founded.

10° On the other hand, the diagram

$$\begin{array}{ccc} & H_\theta & \\ & \nwarrow l & \\ k \uparrow & & \\ M & \xrightarrow{j_{G_k}} & \text{Ult}(M, G_k) \end{array}$$

can be completed by setting $l([f]_{G_k}) := k(f)(k[\cup \text{dom } f])$.

(NB. It is a matter of routine to verify that l is correctly defined, elementary, and that $l \circ j_{G_k} = k$.)

11° This is a contradiction.

□

XV. Proposition. Let δ be Woodin and let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$. Then $\text{Ult}(V, G)$ is closed for ω -sequences.

Proof.

1° Suppose that $\dot{f} : \omega \rightarrow \text{Ult}(V, \text{gen})$. For a given $a_0 \in \mathbb{Q}_\delta$, we will find $a \leq a_0$ and $g : a \rightarrow V$ such that $a \Vdash \dot{f} = [g]$. By a density argument, this will yield the conclusion.

2° For every $n < \omega$, there is a maximal anti-chain A_n such that

$$(\forall b \in A_n)(\exists h : b \rightarrow V)(b \Vdash \dot{f} = [h]).$$

3° For $n < \omega$ and $b \in A_n$, let us fix a function $h_{n,b}$ witnessing the above statement.

4° By semi-proper reflection, there is an inaccessible $\kappa < \delta$ with $a_0 \in V_\kappa$ such that every $A_n \cap \mathbb{Q}_\kappa$ is semi-proper in $\mathbb{Q}_\kappa$.

5° Let a consist of all countable $M \prec H(\kappa^+)$ such that

- a. $M \cap (\cup a_0) \in a_0$
- b. $(A_n \cap \mathbb{Q}_\kappa : n < \omega) \in M$
- c. M captures $A_n \cap \mathbb{Q}_\kappa$ for every $n < \omega$.

6° **Claim.** $a \in \mathbb{Q}_\delta$

Proof.

- 1' Let F be an algebra on $H(\kappa^+)$. We want to find $M \in a$ closed for F .
- 2' Let $M_0 \prec (H_\delta, \in, \kappa, F, (A_n \cap \mathbb{Q}_\kappa : n < \omega), a_0)$ be countable with $M \cap (\cup a_0) \in a_0$.
- 3' Suppose that $M_n \prec (H_\delta, \in, \kappa, (A_n \cap \mathbb{Q}_\kappa : n < \omega))$ has been defined.
- 4' Since $A_n \cap \mathbb{Q}_\kappa$ is semi-proper, by lemma 3-IV, we can choose $M_{n+1} \prec H_\delta$ such that:
 - a. $M_n \subseteq M_{n+1}$
 - b. $M_{n+1} \cap V_\kappa$ end-extends $M_n \cap V_\kappa$;
 - c. M_{n+1} captures $A_n \cap \mathbb{Q}_\kappa$.
- 5' Let $N := \bigcup_{n < \omega} M_n \prec H_\delta$.
- 6' We have
 - a. $N \cap (\cup a_0) = M_0 \cap (\cup a_0) \in a_0$,
 - b. $(A_n \cap \mathbb{Q}_\kappa : n < \omega) \in N$
 - c. N captures $A_n \cap \mathbb{Q}_\kappa$ for every $n < \omega$.
- 7' Thus, $M := N \cap H(\kappa^+) \in a$.
- 8' Since $F \in N$, model M is closed for F , so it is as required.

□

7° Obviously, $a \leq a_0$.

8° We now work towards defining $g : a \rightarrow V$ such that $a \Vdash \dot{f} = [g]$.

9° **Claim.** For every $M \in a$ and every $n < \omega$, there is a unique $b_{M,n}$ such that

$$b_{M,n} \in M \cap A_n \cap \mathbb{Q}_\kappa, \quad M \cap (\cup b_{M,n}) \in b_{M,n}.$$

Proof.

- 1' There is at least one such $b_{M,n}$ since $M \in a$ captures $A_n \cap \mathbb{Q}_\kappa$ by the definition of a .
- 2' Suppose that there are two distinct such objects b_0 and b_1 .
- 3' Then $b_0 \perp b_1$, so we can set $C := (\cup b_0) \cup (\cup b_1)$ and find $F : C^{<\omega} \rightarrow C$ such that for every $N \in [C]^\omega$ closed for F , either $N \cap (\cup b_0) \notin b_0$ or $N \cap (\cup b_1) \notin b_1$.
- 4' Since $b_0, b_1 \in M$, function F can be chosen in M .
- 5' $N := M \cap C \in [C]^\omega$ is closed for F .
- 6' $N \cap (\cup b_i) = M \cap (\cup b_i) \in b_i$ for $i = 0, 1$.
- 7' Thus, this object N contradicts 9°3'.

□

- 10° We will achieve condition $a \Vdash [g] : \omega \rightarrow V$ by defining g to of type $a \rightarrow (\omega \rightarrow V)$.
- 11° Given $M \in a$ and $n < \omega$, we set $g(M)(n) := h_{n,b_{M,n}}(M \cap (\cup b_{M,n}))$ (where $h_{n,b}$ was chosen in 3°).
- 12° **Claim.** For every $n < \omega$, $a \Vdash [g](n) = \dot{f}(n)$.

Proof.

- 1' Let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ with $a \in G$ and set $f := \dot{f}^G$. We want to check that $[g](n) = f(n)$.
- 2' For every $M \in a$, it holds $b_{M,n} \in M \cap A_n \cap \mathbb{Q}_\kappa$; this means that $M \rightarrow b_{M,n}$ is a choice function on a .
- 3' Thus, there is $a' \in G$ with $a' \subseteq a$ and $b \in A_n \cap \mathbb{Q}_\kappa$ such that for every $M \in a'$, we have $b_{M,n} = b$.
- 4' There is a unique $b' \in G \cap A_n$.
- 5' **Subclaim.** $b = b'$

Proof.

- 1'' Assume otherwise, i.e. $b \perp b'$.
- 2'' Let $C := (\cup b) \cup (\cup b')$ and let $F : C^{<\omega} \rightarrow C$ such that for every $M \in [C]^\omega$ closed for F , either $M \cap (\cup b) \notin b$ or $M \cap (\cup b') \notin b'$.
- 3'' Since $a', b' \in G$, there is $c \in G$ with $c \leq a', b'$.
- 4'' Let $M \prec (H_\delta, \in, F)$ be countable with $M \cap (\cup c) \in c$.
- 5'' Since $c \leq a'$, we have $M \cap H(\kappa^+) \in a'$.
- 6'' Then $b_{M \cap H(\kappa^+), n} = b$, so $M \cap (\cup b) = (M \cap H(\kappa^+)) \cap (\cup b) \in b$ (cf. 9°).
- 7'' Since $c \leq b'$, we have $M \cap (\cup b') \in b'$.
- 8'' Since $F \in M$, we see that $M \cap C$ is closed for F ; by the previous two points, it contradicts the choice of F .

□

- 6' We now have

$$f(n) = [h_{n,b'}] \quad (1)$$

$$= [h_{n,b}] \quad (2)$$

$$= [(h_{n,b}(M) : M \in a')] \quad (3)$$

$$= [(h_{n,b_{n,M}}(M) : M \in a')] \quad (4)$$

$$= [(h_{n,b_{n,M}}(M) : M \in a)] \quad (5)$$

$$= [(g(M)(n) : M \in a)] \quad (6)$$

$$= [g](n), \quad (7)$$

for the following reasons:

- (1) since $b' \in A_n$, we have $b' \Vdash \dot{f}(n) = h_{n,b'}$; we also know that $b' \in G$;

- (2) $b = b'$
- (4) for $M \in a'$, we have $b_{n,M} = b$ by the choice of b ;
- (6) by the definition of g .

□

□

XVI. **Corollary.** Let δ be Woodin and let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$. Then $\omega_1^{V[G]} = \omega_1^{\text{Ult}(V,G)} = \delta$.

Proof.

- 1° By c of 2-XVI and 2-XIV, we have $\delta = j_G(\omega_1^V)$.
- 2° This means $\delta = \omega_1^{\text{Ult}(V,G)}$.
- 3° By the previous proposition, we have $\omega_1^{\text{Ult}(V,G)} = \omega_1^{V[G]}$.
- 4° This concludes the argument.

□

4 Fixing the Theory of Chang's Model

- I. **Lemma.** Let $\delta < \lambda$ be inaccessible, let a_δ consist of all countable self-generic $M \prec H(\delta^+)$, and let $G \rightsquigarrow V^{\mathbb{Q}_\lambda}$. Then $G \cap \mathbb{Q}_\delta \rightsquigarrow V^{\mathbb{Q}_\delta}$ if and only if $a_\delta \in G$.

Proof.

- 1° We first prove $(\Leftarrow)$.
- 2° Assume $a := a_\delta \in G$ and let us show that $G \cap \mathbb{Q}_\delta \rightsquigarrow V^{\mathbb{Q}_\delta}$.
- 3° To that end, consider arbitrary predense $D \subseteq G \cap \mathbb{Q}_\delta$ and let us show that $D \cap G \cap \mathbb{Q}_\delta \neq \emptyset$.
- 4° $b := a \cap \{M \in [H(\delta^+)]^\omega : D \in M\} \in G$
- 5° For $M \in b$, choose $f(M)$ as to witness the fact that M captures D , i.e. such that

$$f(M) \in D \cap M, \quad M \cap \bigcup f(M) \in f(M).$$

- 6° $f : b \rightarrow H(\delta^+)$ is a choice function, so normality yields $c \in G$ with $c \subseteq b$ and $d \in D$ such that for all $M \in c$, we have $f(M) = d$.
- 7° This means that for every $M \in c$, we have $M \cap (\cup d) \in d$, while $\cup d \subseteq H(\delta^+) = \cup c$.
- 8° In conclusion, $c \leq d$.
- 9° Since $c \in G$, we have $d \in G$.
- 10° Thus, $d \in D \cap G \cap \mathbb{Q}_\delta$.
- 11° Let us now prove $(\Rightarrow)$.

- 12° To show $a \in G$, it suffices to show $j[H(\delta^+)] \in j(a)$.
- 13° Obviously, $j[H(\delta^+)] \prec j(H(\delta^+))$.
- 14° $|j[H(\delta^+)]|^{\text{Ult}(V,G)} = |[(M : M \in [H(\delta^+)]^\omega)]|^{\text{Ult}(V,G)} = |[(|M| : M \in [H(\delta^+)]^\omega)]| = |(\omega : M \in [H(\delta^+)]^\omega)| = j(\omega) = \omega$
- 15° It remains to verify, in $\text{Ult}(V, G)$, that $j[H(\delta^+)]$ captures every predense $D \subseteq j(\mathbb{Q}_\delta)$ with $D \in j[H(\delta^+)]$.
- 16° Let D be one such set and let D' witness $D \in j[H(\delta^+)]$, i.e. $D = j(D')$.
- 17° D' is predense in $\mathbb{Q}_\delta$.
- 18° Thus, there is $b \in D' \cap G \cap \mathbb{Q}_\delta$.
- 19° $j(b) \in j(D') \cap j[H(\delta^+)] = D \cap j[H(\delta^+)]$
- 20° Also, $j[H(\delta^+)] \cap j(\cup b) = j[H(\delta^+)] \cap j(\cup b) = j[\cup b] \in j(b)$ (the last equality being due to the fact that $b \in G$).
- 21° Thus, $j(b)$ is as required.

□

II. **Lemma.** Let δ be a Woodin limit of Woodins and let $G \rightsquigarrow V^{\mathbb{Q}_\delta}$. Then:

- a. Every $x \in \text{Ord}^\omega \cap V[G]$ is $<\delta$ -generic⁷ over V ;
- b. $\delta = \sup\{\omega_1^{V[x]} : x \in \text{Ord}^\omega \cap V[G]\}$.

Proof.

Part II.a.

- 1° Recall that, in V , for every $\lambda < \delta$ Woodin, set a_λ is projectively stationary (see 3-XIII).
- 2° Thus, $(\forall \alpha < \delta)(\forall b \in \mathbb{Q}_\delta)(\exists \lambda \in (\alpha, \delta))(a_\lambda \parallel b)$.
- 3° We now work in $V[G]$.
- 4° By 2°, set $A := \{\lambda < \delta : a_\lambda \in G\}$ is cofinal in δ .
- 5° Let $x : \omega \rightarrow \text{Ord}$ be arbitrary, named by some $\dot{x}$.
- 6° For each $i < \omega$, there is a maximal anti-chain $A_i \subseteq \mathbb{Q}_\delta$ such that each element of A_i decides $\dot{x}(i)$.
- 7° Let b_i be the unique element of $A_i \cap G$ ($i < \omega$).
- 8° $\text{rank}(b_i) < \delta = \omega_1^{V[G]}$ for every $i < \omega$.
- 9° Thus, there is $\lambda \in A$ such that $b_i \in \mathbb{Q}_\lambda^V$ for every $i < \omega$.
- 10° In other words, $A_i \cap G \cap \mathbb{Q}_\lambda^V \neq \emptyset$ for every $i < \omega$.
- 11° Then $x \in V[G \cap \mathbb{Q}_\lambda]$, as it can be defined there by

$$x(i) = n \iff (\exists p \in A_i \cap G \cap \mathbb{Q}_\lambda^V)(p \Vdash_{\mathbb{Q}_\delta}^V \dot{x}(i) = n).$$

⁷This means that there is some $\mathbb{P} \in V$ with $|\mathbb{P}|^V < \delta$ and $g \rightsquigarrow V^{\mathbb{P}}$ such that $V[x] = V[g]$.

12° Since $a_\lambda \in G$, the previous lemma implies that $G \cap \mathbb{Q}_\lambda \rightsquigarrow V^{\mathbb{Q}_\lambda}$.

13° We can now conclude that x is $<\delta$ -generic over V .

Part II.b.

14° $\delta = \omega_1^{V[G]} \geq \omega_1^{V[x]}$ for every $x \in \text{Ord}^\omega \cap V[G]$.

15° On the other hand, for every $\alpha < \delta = \omega_1^{V[G]}$, there is a real x coding a well-order on ω of order-type α .

16° Thus, $\alpha < \omega_1^{V[x]}$.

17° We conclude $\delta = \sup\{\omega_1^{V[x]} : x \in \text{Ord}^\omega \cap V[G]\}$.

□

III. Main Lemma. Let us work in $V^{\text{Col}(\omega, \Omega)}$, where Ω is large enough. Suppose that δ is a Woodin limit of Woodins in V . Then for every $H \rightsquigarrow V^{\text{Col}(\omega, <\delta)}$, there is some $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ such that

$$\text{Ord}^\omega \cap V[G] = \text{Ord}^\omega \cap V[H].$$

Proof.

We first show that there are some $G_0 \rightsquigarrow V^{\mathbb{Q}_\delta}$ and $H_0 \rightsquigarrow V^{\text{Col}(\omega, <\delta)}$ such that $\text{Ord}^\omega \cap V[G_0] = \text{Ord}^\omega \cap V[H_0]$.

1° Let $G_0 \rightsquigarrow V^{\mathbb{Q}_\delta}$ be arbitrary.

2° Let $\mathbb{S}$ be the set of all $g \rightsquigarrow V^{\text{Col}(\omega, <\alpha)}$, $\alpha < \delta$, that belong to some $V[x]$, $x \in \text{Ord}^\omega \cap V[G_0]$. Let this set be ordered by the reverse inclusion.

3° Let $K \rightsquigarrow V[G_0]^\mathbb{S}$ and let $H_0 := \bigcup K$.

4° **Claim.** $H_0 \rightsquigarrow V^{\text{Col}(\omega, <\delta)}$

Proof.

1' It is clear that H_0 is a filter, so we show that it is V -generic.

2' Let $D \subseteq \text{Col}(\omega, <\delta)$ be arbitrary dense set. We want to show that $H_0 \cap D \neq \emptyset$.

3' Let $D' := \{g \in \mathbb{S} : g \cap D \neq \emptyset\}$.

4' **Subclaim.** D' is dense in $\mathbb{S}$.

Proof.

1'' Let $g \in \mathbb{S}$ be arbitrary. We want to find $h \leq g$ with $h \in D'$.

2'' Let $\alpha < \delta$ and $x \in \text{Ord}^\omega \cap V[G_0]$ be such that $g \rightsquigarrow V^{\text{Col}(\omega, <\alpha)}$ and $g \in V[x]$.

3'' Since D is dense in $\text{Col}(\omega, <\delta)$ and $g \rightsquigarrow V^{\text{Col}(\omega, <\alpha)}$, there is $p \in D$ such that $p \restriction (\omega \times \alpha) \in g$.

4'' Let $\beta \in (\alpha, \delta)$ be such that $p \in \text{Col}(\omega, <\beta)$.

5'' Since δ is inaccessible and x is $<\delta$ -generic, we have

$$|P(\mathbb{B}(\text{Col}(\omega, <\beta))^V)|^{V[x]} < \delta.$$

6'' By II.b, there is $y \in \text{Ord}^\omega \cap V[G_0]$ such that

$$|P(\mathbb{B}(\text{Col}(\omega, <\beta))^V)|^{V[x]} < \omega_1^{V[y]}.$$

7'' Let $z := x \oplus y \in \text{Ord}^\omega \cap V[G_0]$.

8'' Let $\mathbb{Q}$ be the canonical poset such that

$$\text{Col}(\omega, <\beta) \cong \text{Col}(\omega, <\alpha) \times \mathbb{Q}.$$

9'' In $V[z]$, we have that $P(\mathbb{Q})^{V[x]}$ is countable (cf. 4°4'6'').

10'' Thus, in $V[z]$, there is $g' \rightsquigarrow V[x]^\mathbb{Q}$ such $g' \ni p - (p \restriction (\omega \times \alpha))$.

11'' Let $h := g \times g' \rightsquigarrow V^{\text{Col}(\omega, <\beta)}$.

12'' We have $h \in V[z]$, so $h \in \mathbb{S}$ and $h \leq g$.

13'' Also, $h \ni p$, so $h \cap D \neq \emptyset$ and consequently, $h \in D'$. □

5' Let $g \in D' \cap K$. Then $g \cap D \neq \emptyset$, while $g \subseteq \bigcup K = H_0$. □

5° **Claim.** $\text{Ord}^\omega \cap V[G_0] \subseteq \text{Ord}^\omega \cap V[H_0]$

Proof.

1' Let $x \in \text{Ord}^\omega \cap V[G_0]$ be arbitrary.

2' Let $D_x := \{g \in \mathbb{P} : x \in V[g]\}$.

3' **Subclaim.** D_x is dense in $\mathbb{S}$.

Proof.

1'' Let $g \in \mathbb{S}$ be arbitrary. We want to find $h \in D_x$ with $h \supseteq g$.

2'' Let $\alpha < \delta$ and $y \in \text{Ord}^\omega \cap V[G_0]$ be such that $g \rightsquigarrow V^{\text{Col}(\omega, <\alpha)}$ and $g \in V[y]$.

3'' For h , we need to know:

i. $\beta \in (\alpha, \delta)$ such that $h \rightsquigarrow V^{\text{Col}(\omega, <\beta)}$;

ii. $w \in \text{Ord}^\omega \cap V[G_0]$ such that $h \in V[w]$;

and we need to ensure:

iii. $x \in V[h]$

iv. $h \cap \text{Col}(\omega, <\alpha) = g$.

4'' Let $z := x \oplus y \in \text{Ord}^\omega \cap V[G_0]$. We will ensure $z \in V[h]$, which will suffice for 5°3'3''iii.

5'' Note also that $g \in V[z]$.

6'' Since z is $<\delta$ -generic over V , it is $<\delta$ -generic over $V[g]$ as well. More precisely, there are $\mathbb{Q} \in V^{\text{Col}(\omega, <\alpha)}$ and g' such that:

- a. $V^{\text{Col}(\omega, < \alpha)} \models \ulcorner \dot{\mathbb{Q}} \text{ is a poset } \urcorner$;
 - b. $\text{rank}(\dot{\mathbb{Q}}) < \delta$
 - c. $g' \in V[z]$
 - d. $g' \rightsquigarrow V[g]^{\mathbb{Q}_g}$
 - e. $V[z] = V[g][g']$.
- 7'' Now, there is $\beta < \delta$ and a complete embedding

$$k : \text{Col}(\omega, < \alpha) * \dot{\mathbb{Q}} \rightarrow \mathbb{B}(\text{Col}(\omega, < \beta))$$

such that $k(p) = p$ for $p \in \text{Col}(\omega, < \alpha)$. (Existence of such k follows from lemma 26.9. of [1].)

- 8'' Note that $|P(\mathbb{B}(\text{Col}(\omega, < \beta)))|^{V[z]} < \delta$.
- 9'' By II.b, there is some $w' \in \text{Ord}^\omega \cap V[G_0]$ such that

$$|P(\mathbb{B}(\text{Col}(\omega, < \beta)))|^{V[z]} < \omega_1^{V[w']}.$$

- 10'' Let $w := z \oplus w' \in \text{Ord}^\omega \cap V[G_0]$.
- 11'' Then in $V[w]$, there is $h \rightsquigarrow V^{\text{Col}(\omega, < \beta)}$ such that $k^{-1}[h] = g * g'$.
- 12'' h is as required in 5°3'3''.

□

- 4' Thus, there is $g \in K \cap D_x$.
- 5' This means $x \in V[g]$.
- 6' Since $g \in K$, there is $\alpha < \delta$ such that $g = H_0 \restriction \alpha$.
- 7' Then $x \in V[g] \subseteq V[H_0]$.

□

6° **Claim.** $\text{Ord}^\omega \cap V[G_0] \supseteq \text{Ord}^\omega \cap V[H_0]$

Proof.

- 1' Let $x \in \text{Ord}^\omega \cap V[H_0]$ be arbitrary.
- 2' Since δ is inaccessible in V and x is an ω -sequence of ordinals, there is $\alpha < \delta$ such that $x \in V[H_0 \restriction \alpha]$.
- 3' Then $g := H_0 \restriction \alpha \in \mathbb{S}$, so there is $y \in \text{Ord}^\omega \cap V[G_0]$ such that $g \in V[y]$.
- 4' We now have $x \in V[g] \subseteq V[y] \subseteq V[G_0]$.

□

7° Thus, we have verified that $\text{Ord}^\omega \cap V[G_0] = \text{Ord}^\omega \cap V[H_0]$.

We now show that for a given H , we can find $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ such that $\text{Ord}^\omega \cap V[G] = \text{Ord}^\omega \cap V[H]$.

- 8° Let $\Gamma \rightsquigarrow V^{\text{Col}(\omega, \Omega)}$ be arbitrary. We work in $V[\Gamma]$.
- 9° Let $H \rightsquigarrow V^{\text{Col}(\omega, < \delta)}$ be arbitrary. We want to find $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ such that $\text{Ord}^\omega \cap V[G] = \text{Ord}^\omega \cap V[H]$.

- 10° Consider the sentence $\phi(\delta, H)$: “there is a generic $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ such that $\text{Ord}^\omega \cap V[G] = \text{Ord}^\omega \cap V[H]$ ”. We need to show $V[\Gamma] \models \phi(\delta, H)$.
- 11° Let $G_0, H_0 \in V[\Gamma]$ be as above. There is some $\Delta \rightsquigarrow V[H_0]^{\text{Col}(\omega, \Omega)}$ such that $V[\Gamma] = V[H_0][\Delta]$.
- 12° $V[H_0][\Delta] \models \phi(\delta, H_0)$
- 13° There is some $p \in \text{Col}(\omega, \Omega)$ such that

$$V[H_0] \models (p \Vdash \phi(\delta, H_0)).$$

- 14° Since $\text{Col}(\omega, \Omega)$ is homogeneous, we have

$$V[H_0] \models (\text{Col}(\omega, \Omega) \Vdash \phi(\delta, H_0)).$$

- 15° This means that for some $q \in \text{Col}(\omega, < \delta)$ and the canonical name **gen** for the $\text{Col}(\omega, < \delta)$ -generic, we have

$$V \models (q \Vdash (\text{Col}(\omega, \Omega) \Vdash \phi(\delta, \text{gen}))).$$

- 16° However, in V , we have

$$\text{Col}(\omega, < \delta) \Vdash \phi(\delta, \text{gen}) \Leftrightarrow \phi(\delta, \pi_* \text{gen})$$

for every $\pi \in \text{Aut}(\mathbb{B}(\text{Col}(\omega, < \delta)))$, so the homogeneity of $\text{Col}(\omega, < \delta)$ implies

$$\text{Col}(\omega, < \delta) \Vdash (\text{Col}(\omega, \Omega) \Vdash \phi(\delta, \text{gen})).$$

- 17° Thus,

$$V[H] \models (\text{Col}(\omega, \Omega) \Vdash \phi(\delta, H)).$$

- 18° There is $\Delta' \rightsquigarrow V[H]^{\text{Col}(\omega, \Omega)}$ such that $V[\Gamma] = V[H][\Delta']$. We have

$$V[\Gamma] = V[H][\Delta'] \models \phi(\delta, H),$$

which was to be proved. □

IV. **Definition.** *Chang’s model* $\mathbb{C}$ is the smallest inner model for ZF that is closed for ω -sequences. Equivalently, it is the smallest inner model for ZF that contains all ω -sequences of ordinals, i.e. $\mathbb{C} := L(\text{Ord}^\omega)$.

V. **Theorem.** Let us work in $V^{\text{Col}(\omega, \Omega)}$, where Ω is large enough. Suppose that δ is a Woodin limit of Woodins in V and let $g \rightsquigarrow V^{\text{Col}(\omega, < \delta)}$. Then there is an elementary embedding $j : \mathbb{C}^V \prec \mathbb{C}^{V[g]}$.

Proof.

1° By 3-XIII yields a generic $G \rightsquigarrow V^{\mathbb{Q}_\delta}$ such that

$$\text{Ord}^\omega \cap V[G] = \text{Ord}^\omega \cap V[g].$$

2° Let $j_G : V \prec \text{Ult}(V, G) =: M$ be the generic embedding.

3° We have

$$j := j_G \restriction \mathbb{C}^V : \mathbb{C}^V \prec \mathbb{C}^M.$$

4° Since M is ω -closed in $V[G]$, we have $\text{Ord}^\omega \cap M = \text{Ord}^\omega \cap V[G]$, i.e. $\mathbb{C}^M = \mathbb{C}^{V[G]}$.

5° By 1°, we also have $\mathbb{C}^{V[G]} = \mathbb{C}^{V[g]}$.

6° Thus, $j : \mathbb{C}^V \prec \mathbb{C}^{V[g]}$.

□

VI. **Theorem.** Let δ be a Woodin limit of Woodins and let G be $<\delta$ -generic over V . Then $\mathbb{C}^V \equiv_{\mathbb{R}^V} \mathbb{C}^{V[G]}$.

Proof.

1° Formally, we need to prove that for every $\mathbb{P} \in V_\delta$, we have

$$\mathbb{C} \models \phi(a) \iff V^\mathbb{P} \models (\mathbb{C} \models \phi(a))$$

for all formulas ϕ and all reals a .

2° Let Ω be large enough and let $\Gamma \rightsquigarrow V^{\text{Col}(\omega, \Omega)}$. We work in $V[\Gamma]$.

3° For the conclusion, it suffices to show

$$\mathbb{C}^V \equiv_{\mathbb{R}^V} \mathbb{C}^{V[G]}$$

for every $G \rightsquigarrow V^\mathbb{P}$.

4° Fix some $G \rightsquigarrow V^\mathbb{P}$ and let $g \rightsquigarrow V[G]^{\text{Col}(\omega, <\delta)}$.

5° By the previous theorem, there is an embedding $j : \mathbb{C}^{V[G]} \prec \mathbb{C}^{V[G][g]}$.

6° Since forcings $\mathbb{P} \times \text{Col}(\omega, <\delta)$ and $\text{Col}(\omega, <\delta)$ are equivalent in V , there is $h \rightsquigarrow V^{\text{Col}(\omega, <\delta)}$ such that $V[G][g] = V[h]$.

7° By the previous theorem, there is an embedding $k : \mathbb{C}^V \prec \mathbb{C}^{V[h]} = \mathbb{C}^{V[G][g]}$.

8° After noting that both j and k fix reals, we come to the conclusion by referencing 5° and 7°.

□

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